

Classification Methods

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Outline

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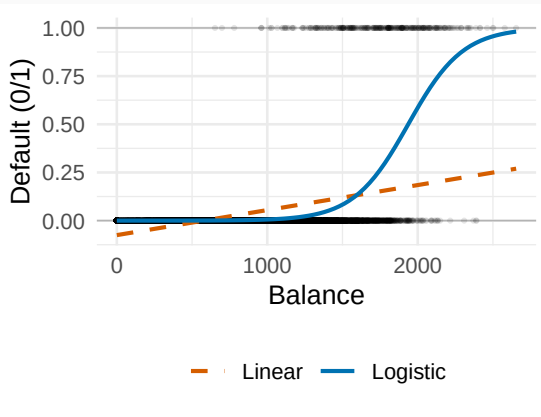
Classification Problems and Decision Boundaries

The Classification Problem

- The response $y^{(i)}$ is qualitative: it takes one of K classes, $y^{(i)} \in \{1, \dots, K\}$, or $y^{(i)} \in \{0, 1\}$ when $K = 2$.
- Training data: $\{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^n$ with feature vectors $\mathbf{x}^{(i)} = (x_1^{(i)}, \dots, x_p^{(i)})^\top \in \mathbb{R}^p$.
- Goal: learn a classifier $f : \mathbb{R}^p \rightarrow \{1, \dots, K\}$ such that $\hat{y}_0 = f(\mathbf{x}_0)$ is correct for a new observation $\mathbf{x}_0$.
- Examples: spam vs. not spam, loan default, land-cover class from spectral bands, disease present vs. absent.
- Regression predicts a quantitative response; classification predicts a category, usually through an estimated probability for each class.

Why Not Linear Regression?

- Coding a binary response as $y^{(i)} \in \{0, 1\}$ and fitting $\hat{y} = \hat{\beta}_0 + \hat{\beta}^\top \mathbf{x}$ gives fitted values that are not confined to $[0, 1]$, so they are not valid probabilities.
- For $K > 2$, coding the classes as $1, 2, 3, \dots$ imposes an ordering and equal spacing that the classes do not have.
- We need a model whose output always lies in $[0, 1]$.



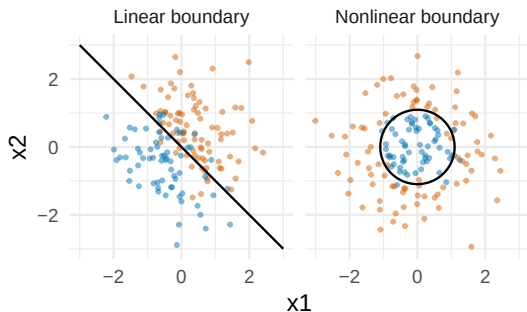
Default data: the linear fit is negative for small balances, the logistic fit stays in $[0, 1]$.

Classifiers Estimate Class Probabilities

- Conditional class probability: $p_k(\mathbf{x}) = P(y = k \mid \mathbf{x})$ for $k = 1, \dots, K$, with $\sum_{k=1}^K p_k(\mathbf{x}) = 1$.
- Bayes classifier: assign $\mathbf{x}$ to the most probable class, $f^*(\mathbf{x}) = \arg \max_k p_k(\mathbf{x})$.
- Error rate on n observations: $\frac{1}{n} \sum_{i=1}^n \mathbb{1}\{y^{(i)} \neq \hat{y}^{(i)}\}$.
- The Bayes classifier has the smallest possible expected error rate, the Bayes error rate $1 - \mathbb{E}_{\mathbf{x}}[\max_k p_k(\mathbf{x})]$, but it cannot be used in practice because $p_k(\mathbf{x})$ is unknown.
- Practical classifiers approximate $p_k(\mathbf{x})$: logistic regression with a parametric function, k-NN with class proportions among nearby training points.

Decision Boundaries

- A classifier splits the feature space $\mathbb{R}^p$ into regions $R_k = \{\mathbf{x} : f(\mathbf{x}) = k\}$; the decision boundary is where regions meet.
- For two classes, the Bayes decision boundary is $\{\mathbf{x} : p_1(\mathbf{x}) = 0.5\}$.
- A linear boundary is a hyperplane, $\beta_0 + \beta^\top \mathbf{x} = 0$; a nonlinear boundary is curved.



Classification Methods

Logistic Regression

The Logistic Regression Model

- Binary response $y^{(i)} \in \{0, 1\}$; model the probability $p(\mathbf{x}) = P(y = 1 \mid \mathbf{x})$.
- Pass a linear score through the logistic (sigmoid) function $\sigma(z) = 1/(1 + e^{-z})$:

$$p(\mathbf{x}) = \sigma(\beta_0 + \boldsymbol{\beta}^\top \mathbf{x}) = \frac{1}{1 + e^{-(\beta_0 + \boldsymbol{\beta}^\top \mathbf{x})}}, \quad \boldsymbol{\beta} = (\beta_1, \dots, \beta_p)^\top.$$

- Properties: $\sigma(z) \in (0, 1)$, $\sigma(0) = 0.5$, σ is increasing, and $\sigma(-z) = 1 - \sigma(z)$.
- Fitted probabilities $\hat{p}(\mathbf{x})$ are always valid, and the fitted curve is S-shaped rather than a straight line.

Odds and Log-Odds

- Odds: $\frac{p(\mathbf{x})}{1 - p(\mathbf{x})} \in (0, \infty)$; for example $p = 0.8$ gives odds of 4 (four to one in favour of $y = 1$).
- Rearranging the model, the log-odds (logit) is linear in $\mathbf{x}$:

$$\log \frac{p(\mathbf{x})}{1 - p(\mathbf{x})} = \beta_0 + \boldsymbol{\beta}^\top \mathbf{x}.$$

- Increasing x_j by one unit with the other features fixed adds β_j to the log-odds, i.e. multiplies the odds by e^{β_j} .
- The change in $p(\mathbf{x})$ itself is not constant: it depends on where on the S-curve $\mathbf{x}$ currently sits.

Estimation: Maximum Likelihood

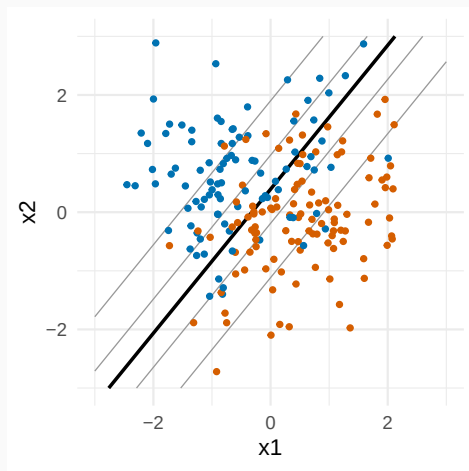
- Model $y^{(i)} \mid \mathbf{x}^{(i)} \sim \text{Bernoulli}(p^{(i)})$ with $p^{(i)} = p(\mathbf{x}^{(i)})$, observations independent.
- Log-likelihood:

$$\ell(\beta_0, \boldsymbol{\beta}) = \sum_{i=1}^n \left[y^{(i)} \log p^{(i)} + (1 - y^{(i)}) \log (1 - p^{(i)}) \right].$$

- Maximizing ℓ is the same as minimizing the cross-entropy loss $-\ell/n$.
- No closed-form solution, unlike least squares: the maximum is found iteratively by Newton–Raphson (IRLS), which is what `glm()` in R uses.
- Let $\tilde{\mathbf{x}}^{(i)} = (1, \mathbf{x}^{(i)\top})^\top$, $\tilde{\boldsymbol{\beta}} = (\beta_0, \boldsymbol{\beta}^\top)^\top$, and $\tilde{\mathbf{X}} \in \mathbb{R}^{n \times (p+1)}$ with rows $\tilde{\mathbf{x}}^{(i)\top}$. The gradient is $\nabla \ell(\tilde{\boldsymbol{\beta}}) = \tilde{\mathbf{X}}^\top (\mathbf{y} - \mathbf{p})$, with $\mathbf{y} = (y^{(i)})_{i=1}^n$ and $\mathbf{p} = (p^{(i)})_{i=1}^n$.

Predictions and the Decision Boundary

- Predict $\hat{y} = 1$ if $\hat{p}(\mathbf{x}) > \tau$, else $\hat{y} = 0$; the default is $\tau = 0.5$.
- For $\tau = 0.5$, $\hat{y} = 1 \iff \hat{\beta}_0 + \hat{\beta}^\top \mathbf{x} > 0$, so the boundary $\hat{\beta}_0 + \hat{\beta}^\top \mathbf{x} = 0$ is a hyperplane.
- Logistic regression is a linear classifier, although $p(\mathbf{x})$ is nonlinear in $\mathbf{x}$.
- Another τ gives $\hat{\beta}_0 + \hat{\beta}^\top \mathbf{x} = \log \frac{\tau}{1-\tau}$, a parallel hyperplane.



Black: $\hat{p} = 0.5$; grey: 0.1, 0.3, 0.7, 0.9.

Worked Example: Default Data

- Default: 10,000 credit card customers, response default (333 defaults), predictors balance, income, student.
- Fit: `glm(default ~ balance, data = Default, family = binomial)`.

	Estimate	Std. error	<i>z</i>	<i>p</i> -value
Intercept	-10.6513	0.3612	-29.49	$< 2 \times 10^{-16}$
balance	0.00550	0.00022	24.95	$< 2 \times 10^{-16}$

- $e^{100\hat{\beta}_1} \approx 1.73$: each additional \$100 of balance multiplies the odds of default by about 1.73 (95% CI [1.66, 1.81]).
- Predicted default probability: $\hat{p}(\text{balance} = 1000) \approx 0.006$ and $\hat{p}(\text{balance} = 2000) \approx 0.586$.

Multiple Logistic Regression and Confounding

- Additional predictors enter the log-odds linearly, $\log \frac{p(\mathbf{x})}{1-p(\mathbf{x})} = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p$, and each β_j is the effect of x_j on the log-odds holding the other predictors fixed.
- A categorical predictor enters through dummy variables, for example $x = 1$ for a student and $x = 0$ otherwise.
- In `Default`, the student coefficient is $+0.405$ ($p < 0.001$) when student is the only predictor, but -0.647 ($p = 0.006$) once balance and income are included.
- Students carry higher balances on average (about \$988 vs. \$772) and higher balances raise default risk, so students default more overall but less than non-students with the same balance; as in multiple linear regression, the marginal and the conditional effect can differ in sign.

Multinomial Regression

- For $K > 2$ classes, model every class probability with the softmax function, $P(y = k \mid \mathbf{x}) = \exp(\beta_{k0} + \boldsymbol{\beta}_k^\top \mathbf{x}) / \sum_{l=1}^K \exp(\beta_{l0} + \boldsymbol{\beta}_l^\top \mathbf{x})$ for $k = 1, \dots, K$.
- Identifiability: a model is identifiable if distinct parameters always give distinct probabilities. This one is not: adding any constant vector to every $(\beta_{k0}, \boldsymbol{\beta}_k)$ leaves all K probabilities unchanged, since the shift cancels between numerator and denominator.
- Fix a reference class, usually K , by setting $\beta_{K0} = 0, \boldsymbol{\beta}_K = \mathbf{0}$; this pins down a unique parameterization for the remaining $K - 1$ classes.
- Interpretation: relative to the reference class, $\log [P(y = k \mid \mathbf{x}) / P(y = K \mid \mathbf{x})] = \beta_{k0} + \boldsymbol{\beta}_k^\top \mathbf{x}$, so β_{kj} is the change in the log-odds of class k vs. class K per unit increase in x_j , and $e^{\beta_{kj}}$ is the multiplicative change in that odds ratio.

Nonlinear Decision Boundaries

- Add transformed features such as x_1^2 , x_2^2 , x_1x_2 to the linear score; the model stays linear in the parameters, exactly as in polynomial regression.
- Example: the circle $x_1^2 + x_2^2 = r^2$ is a linear boundary in the features (x_1^2, x_2^2) .
- More generally, replace $\mathbf{x}$ with a feature map $\phi(\mathbf{x})$ and fit a linear boundary in $\phi(\mathbf{x})$ -space; any boundary that is linear in some feature map can be captured this way.
- A richer feature map fits the training boundary more closely but risks overfitting; how many terms to add is the same trade-off as choosing polynomial degree.

Classification Methods

k - Nearest Neighbors

k -Nearest Neighbors Classifier

- k -NN is nonparametric: there is no model to fit, and predictions use the training data directly.
- For a new point $\mathbf{x}_0$ and a chosen integer k , let $\mathcal{N}_0$ be the set of indices of the k training points closest to $\mathbf{x}_0$.
- Estimate the class probabilities by the class proportions in the neighbourhood:

$$\hat{p}_j(\mathbf{x}_0) = \frac{1}{k} \sum_{i \in \mathcal{N}_0} \mathbb{1}\{y^{(i)} = j\}.$$

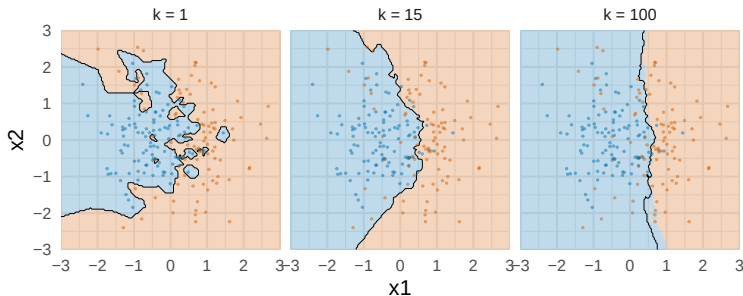
- Classify by majority vote, $\hat{y}_0 = \arg \max_j \hat{p}_j(\mathbf{x}_0)$, with ties broken at random.
- Example: with $k = 3$ and two of the three nearest neighbours in class 1, $\hat{p}_1(\mathbf{x}_0) = 2/3$ and $\hat{y}_0 = 1$.

Distance and Feature Scaling

- Closeness is measured by a distance; the default is Euclidean, $d(\mathbf{a}, \mathbf{b}) = \|\mathbf{a} - \mathbf{b}\|_2 = \sqrt{\sum_{j=1}^p (a_j - b_j)^2}$.
- Alternatives: Manhattan, $\sum_{j=1}^p |a_j - b_j|$, and the Minkowski family $(\sum_{j=1}^p |a_j - b_j|^q)^{1/q}$, which contains both ($q = 1$ and $q = 2$).
- Features with a large spread dominate the distance: in Default, income has a standard deviation of about \$13,300 and balance about \$480, so raw distances are driven almost entirely by income, the weaker predictor of default.
- Standardize before computing distances, $z_j^{(i)} = (x_j^{(i)} - \bar{x}_j)/s_j$, where $\bar{x}_j$ and s_j are the sample mean and standard deviation of feature j ; preprocessing is treated in detail in a later lecture.

The Effect of k

- Small k ($k = 1$): jagged boundary following individual points; training error is 0, which says nothing about new data.
- Large k : smooth, rigid boundary that can miss genuine curvature; $k = n$ predicts the majority class everywhere.
- k is a tuning parameter; choosing it needs held-out data (a later lecture).



k-NN: Strengths and Weaknesses

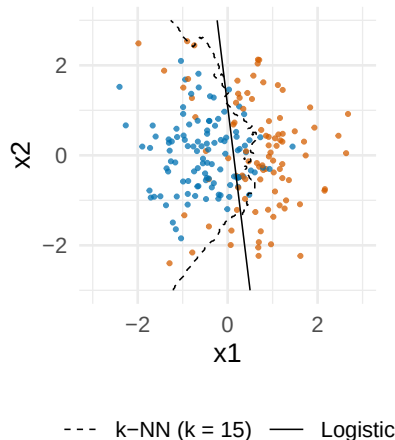
- Strengths: no fitting step, no assumption about the shape of the boundary, a single tuning parameter k , easy to explain.
- Cost: every prediction computes distances to all n training points, $O(np)$ per query, and the whole training set must be stored.
- Curse of dimensionality: as p grows the data become sparse and the “nearest” neighbours are no longer near; irrelevant features add noise to every distance.
- No coefficients: k-NN gives predictions but no summary of how each feature affects the outcome, unlike e^{β_j} in logistic regression.
- Sensitive to feature scaling, as on the previous slide.

Classification Methods

Comparison and Summary

Logistic Regression vs. k -NN on the Same Data

- Same simulated data (curved true boundary); boundaries drawn at $\hat{p} = 0.5$.
- Logistic regression: a straight line, misses the curvature.
- k -NN ($k = 15$): follows the curve, locally jagged.
- Which is better depends on the true boundary and the sample size; judging it needs held-out data.



Logistic Regression vs. k -NN

	Logistic regression	k -NN
Type	Parametric: β_0, β	Nonparametric: stores the training data
Decision boundary	Linear, unless features are added	Flexible, controlled by k
Fitting	Iterative maximum likelihood	None
Cost per prediction	$O(p)$	$O(np)$
Interpretation	Odds ratios e^{β_j} , z -tests, confidence intervals	No coefficients
Main weakness	Misses nonlinear structure unless features are engineered	Feature scaling, irrelevant features, high dimensions

Summary

- Classification predicts a category by estimating the class probabilities $p_k(\mathbf{x})$; the Bayes classifier is the target, and the decision boundary partitions the feature space into class regions.
- Logistic regression: log-odds linear in $\mathbf{x}$, coefficients fitted by maximum likelihood, linear boundary, coefficients read as odds ratios; extends to $K > 2$ classes and to nonlinear boundaries through added features.
- k-NN: majority vote among the k closest training points, flexible boundary controlled by k , needs feature scaling, degrades in high dimensions.
- Still open: how to choose k and how to compare classifiers on data not used for fitting.

Questions?
